

Ekaterina Prytkova, PhD

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📍 Right to work: EU Settlement Scheme – Pre-Settled Status

PROFESSIONAL EXPERIENCE

Postdoctoral Research Fellow in Economics of AI and Data Science

Jan 2024 – Present

University Côte d'Azur & CNRS

Nice, France

- Developed robust, scalable web-scraping and API pipelines to build custom/unique database for analysis of firm's strategic behavior in AI model market.
- Developed an LLM-based pipeline to extract and embed skills from large-scale multilingual job ads to map human skills to an agentic skill framework (OpenClaw), identifying areas of intensifying human–AI task competition.
- Founded and operated a multidisciplinary AI seminar series, organizing expert talks on various AI-focused topics and managing the series' media presence to drive evidence-based discussion on AI development and impact; reached ~1.5k subscribers and average of 60 attendees per event over 5 seasons.
- Conducted QA for PhD research projects, including code review and testing, data pipeline validation, evaluation of methodological fit (modeling and identification strategy) to the problem definition, and project delivery support.

Postdoctoral Research Fellow in Economics of Innovation and Data Science

Sep 2021 – Nov 2023

PILLARS Project, University of Sussex Business School, SPRU

Brighton, United Kingdom

- Built scalable Python and R pipelines to ingest large text corpora (200K patent texts, 1.8M publications) and produce a key indicator of labour automation leveraging LLMs, ML and NLP techniques.
- Conducted causal inference on the impact of digital automation on the future of work using an instrumental variable shift-share design and longitudinal labour data.
- Developed a novel trend-detection pipeline that uses NLP and unsupervised ML to identify and monitor emerging digital technologies using large text corpora (4.5M publications, 800M citations).
- Led a key work package in a large-scale research project, overseeing data analytics and shaping research design and methodology.
- Transformed complex analytical results into clear, actionable data visualizations and reports for technical and non-technical project partners and stakeholders at the granting policy body, supporting evidence-based policy development.
- Mentored a 3-person team of Data Science Master's research assistants, aligning workflows with project goals and improving delivery consistency through structured coordination and review processes.

PhD Researcher/ Teaching Instructor

Jan 2017 – Sep 2021

Friedrich Schiller University Jena

Jena, Germany

- Refined standard patent document classification using NLP feature extraction and supervised ML models (SVM) in Python to analyse innovation trends in the semiconductor and AI industries.
- Integrated and analysed multiple large-scale datasets (employment, R&D, I-O) in R to investigate digital technologies' impact on labour productivity.
- Developed strong public speaking skills through presenting research at international conferences to expert audiences, communicating methods and findings and incorporating feedback.
- Initiated and co-developed an "Introduction to R" course in response to identified skills gaps among Master's students, co-designing the curriculum and delivering practical sessions as a TA.
- Led the transition of in-person teaching to a fully online format, adapting teaching content and delivery methods and implementing online learning infrastructure.

EDUCATION

Friedrich Schiller University Jena

Jan 2017 – Sep 2021

PhD in Economics (summa cum laude)

Jena, Germany

- Dissertation: A Study on the Nature of ICT and AI and Their Impact on Industrial Trajectories

Higher School of Economics

MSc in Management (with distinction)

Sep 2014 – Jun 2016

St. Petersburg, Russia

- Major: Marketing, Forecasting, Innovation Management

Friedrich Schiller University Jena

Faculty of Economics (Visiting period)

Oct 2015 – Mar 2016

Jena, Germany

- Major: Microeconomics, Quantitative Methods, Innovation

Higher School of Economics

BSc in Economics

Sep 2010 – Jun 2014

St. Petersburg, Russia

- Major: Econometrics, Industrial Organization, Game Theory

SKILLS & COMPETENCES

- **Programming & Scripting:** Python, R; SQL, HTML, CSS
- **Data Science & Analytics:** Pandas, NumPy, Scikit-learn, NLTK, spaCy, TensorFlow
- **Data Visualization:** Matplotlib, Seaborn, Plotly, Streamlit
- **Web automation & Scraping:** Playwright, Selenium, Scrapy, BeautifulSoup
- **Tools:** Git/GitHub, LaTeX

CERTIFICATIONS & TRAINING

Build Your LLM From Scratch

Jan 2026 – Mar 2026

Inria

Nice, France

- Tokenizer, Encoder, Decoder, Instruct (SFT), Reinforcement Learning (PPO, DPO, ORPO, VR), Retrieval-Augmented Generation (RAG)

Summer Institute in Computational Social Science

Jun 2025

University College London & Imperial College London

London, United Kingdom

- Workshops: NLP, Network Analysis, LLM Agents, Generative Agent-Based Modelling

Deep Learning & Machine Learning

Oct 2024 – Mar 2025

University Côte d'Azur

Nice, France

- ML: SVM, Clustering, Tree-based Methods, Ensemble Learning, Confident Learning, Multi-label Classification
- DL: TensorFlow, PyTorch, MLP, Word embeddings, transformers, Interpretability Methods

PROFESSIONAL ENGAGEMENT & SERVICE

Developer, Open Source [TechXposure Database](#)

Semantic similarity; Transformers; Contrastive Learning; Zero-shot Learning; Big Data; Future of Work

Co-Founder & Scientific Organizer, [VOILA! Seminar Series](#)

AI Thought Leadership; Expert Ecosystem Building; AI Governance & Ethics; Societal Impact & Outreach

Mentor, [Women in Economics Institute e.V.](#)

Coaching & Capability Building; Leadership, Empowerment & Inclusion; Professional Development Support

Peer Reviewer, academic journals

Research Policy | Industrial and Corporate Change | Science and Public Policy

SELECTED RESEARCH

The Role of Open-Weight Strategy in the AI Model Market: Firm Outcomes and Competitive Implications

Ongoing research

- Artificial Intelligence; Unsupervised Learning; Network Analysis; Webscraping; Empirical IO; Digital Platforms

The Employment Impact of Emerging Digital Technologies

EU study: [CESifo Working Paper \(2025\)](#); US study: [NBER Working Paper \(2025\)](#)

- NLP; Causal Inference; Text Mining; Big Data; Future of Work; Emerging Digital Technologies
- Best Paper Award at the [2nd CESifo / ifo Junior Workshop on Big Data](#); IMF Paper Award at the [Armenian Economic Association 2024 Annual Meetings](#); Best Paper Award at the [2025 CONCORDI Conference](#)

On the Basis of Brain: Neural-Network-Inspired Changes in General-Purpose Chips

[Industrial and Corporate Change](#), Vol. 31, Issue 4., 2022

- IO Modelling; Patent Data; Technology Foresight; Competition; Semiconductor Industry

LANGUAGES & INTERESTS

- **Languages:** English (fluent), Russian (native), Italian (B1), German (A2)
- **Interests:** Painting, (Astro)Photography & Crafts; Hiking, Paddleboarding, Table Tennis; British History